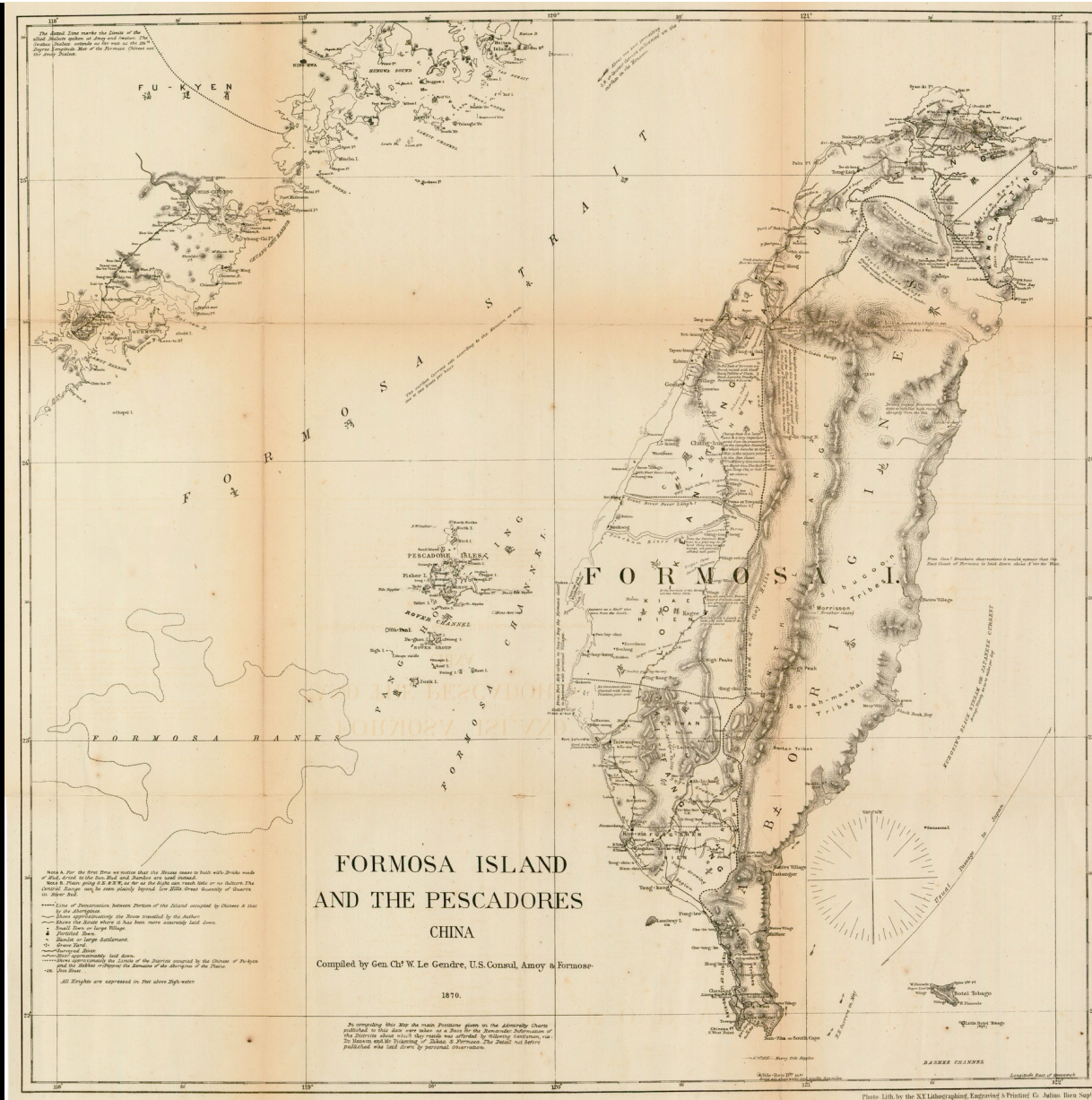


Cartography and Geographic Information System

GEOG 1026
Department of Geography
National Taiwan University

Formosa Island and The Pescadores, China, 1871, Charles W. Le Gendre



Before we start...

Cartography and Geographic Information System

Syllabus

- Ray Chuang raychuang@ntu.edu.tw
- Credit hours: 3 hours
- Prerequisites: N/A
- Lectures and Labs: Fridays 2:20-5:20
- Classroom: Lectures (305) & Labs (305 & computer lab)

TAs

- 介姿淇 Tzu-Chi Chieh
- D14249008@ntu.edu.tw
- 梅立昀 Li-Yun Mei
- r14228011@ntu.edu.tw
- 楊宇農 Yu-Nung Yang
- r14228005@ntu.edu.tw
- Office hour: by appointment

Reference Books

- Chang (2019), Introduction to geographic information systems, McGraw-Hill Higher Education.
- Longley, Goodchild, Maguire, & Rhind (2015), Geographic information science and systems, Wiley.
- DeMers (2011), Fundamentals of geographic information systems, John Wiley & Sons.
- Anthamatten (2020), How to make maps: An introduction to theory and practice of cartography, Routledge.
- Field (2022), Thematic mapping: 101 inspiring ways to visualise empirical data, ESRI Press.
- Monmonier (2018) How to lie with maps, Third edition, University of Chicago Press.

Course Objectives

- Maps are the language of geography. Cartography is the science, technique and art of creating maps. It involves collecting geographic information, selecting and simplifying data, and producing maps that communicate spatial patterns, relationships, and geographic phenomena.
- Geographic information systems (GIS) are computer-based systems to collect, model, process, analyze, and visualize geographic information.
- This course integrates theory and practice by introducing the principles of cartography and the theoretical foundations of GIS, strengthening spatial thinking, and cultivating students' ability to explore geographic issues using geospatial information.

The Real Course Objectives

- You will need the skills of this course for most of the rest courses and the research in this department.
- Therefore, you have to be able to process geographic data and to produce correct, informative maps.
- Also, you have to be able to understand others' maps and extract useful geographic information out of them.

Course Structure in Our Department

地理環境資源學系必修科目表

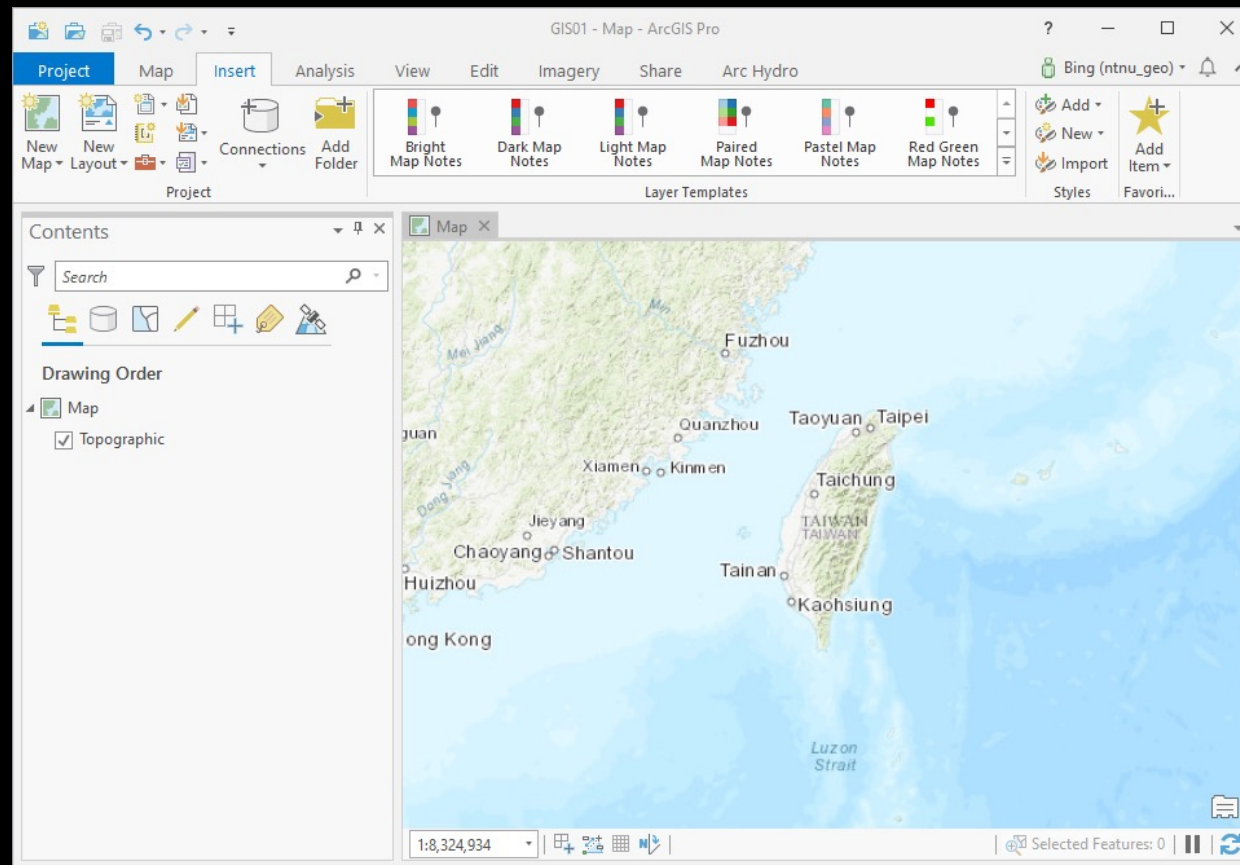
	大 一	大 二	大 三	大 四
上 學 期	微積分 1 (2) 微積分 2 (2) 自然地理學通論 (3) 人文地理學通論 (3)	地形學及實習 (3) 統計學及實習 (4) 城鄉、區域與發展 (3) 遙測學原理 (3)	自然地理調查方法 (3) 人文地理調查方法 (3) 以上二科選一科 環境資源保育經理及實習 (3)	
下 學 期	微積分 3 (2) 程式設計 (3) 地圖與地理資訊系統 (3)	氣候學及實習 (3) 文化、社會與自然 (3) 空間分析 (3)	區域地理 (3)	

115 學年度入學者開始實施。

114 年 5 月 29 日系務會議通過

GIS Software

- ESRI ArcGIS Pro



Schedule

Week	Date	Topics	Map	Lab
1	2/27	No class 228補假		
2	3/6	Introduction 課程簡介	Map history	
3	3/13	Geographic data and information 地理資料與資訊	Map components	Lab 1
4	3/20	Coordinates 坐標系統	Projection	Lab 2
5	3/27	Spatial data type 空間資料類型	Thematic maps	Lab 3
6	4/3	No class 春假		
7	4/10	Attribute data 屬性資料	Advanced thematic maps	Lab 4
8	4/17	Spatial selection 空間選取	Layout	Lab 5

Schedule

Week	Date	Topics	Map	Lab
9	4/24	Midterm 期中考		
10	5/1	No class 勞動節放假		
11	5/8	Spatial classification 空間分類	Coloring	Lab 6
12	5/15	Spatial interpolation 空間內插	Mapping uncertainty	Lab 7
13	5/22	Terrain analysis 地勢分析	Terrain representation	Lab 8
14	5/29	Spatiotemporal GIS 時空資料	Web maps	Lab 9
15	6/5	Final project presentation 期末報告		
16	6/12	Final exam 期末考		

Course schedule may be changed according to the progress.

Important dates

- Exams:
 - 4/24 Midterm
 - 6/12 Final exam
- Final project:
 - 4/17 Grouping
 - 5/8 Proposal
 - 6/5 Presentation
- No class
 - 2/27 National holiday
 - 4/3 Spring break
 - 5/1 National holiday

Grade

- Midterm: 20%
 - Written
 - Hands-on
- Final exam: 25%
 - Written
 - Hands-on
- Quiz, lab & homework: 30%
 - Individual work
- Final project: 25%
 - Team work, peer assessment
- **No cheating, no plagiarism!**

Grade

教育部 函

機關地址：10051臺北市中山南路5號

傳 真：02-23976943

聯絡人：如說明一

電 話：如說明一

受文者：國立臺灣大學

發文日期：中華民國109年8月19日

發文字號：臺教高通字第1090112935號

速別：普通件

密等及解密條件或保密期限：

附件：空白查核表

主旨：為強化各大學校院學位論文品保機制，請各校確依說明辦理，請查照。

Grade

說明：

- 一、依據學位授予法第17條第1項規定，學校授予之學位，有入學資格或修業情形有不實或舞弊情事或論文、作品、成就證明、書面報告、技術報告或專業實務報告有造假、變造、抄襲、由他人代寫或其他舞弊情事之一者，應予撤銷，並公告註銷其已頒給之學位證書；另依專科以上學校學術倫理案件處理原則第6條第1項規定，學校應訂定學術倫理相關規定，包括學術倫理規範、權責單位、修習辦法、違反態樣、處理程序、處分條款及監管機制等規定，並依學校章則訂定程序辦理後，公告周知。為確保學校於接獲涉及學位授予法第17條情事檢舉案件，確實依學校處理程序進行審議，爰嗣後經本部函轉學校涉及學位授予法第17條情事之案件，請於案件審議完後填寫「專科以上學校學生學位論文涉及學位授予法第17條情事查核表

AI Policy

- AI tools may be used as learning aids, but not as substitutes for your own spatial reasoning, GIS workflows, or map design decisions. Students are expected to demonstrate independent understanding of cartographic principles and GIS operations.
- Why does this policy exist?
 - Tool proficiency
 - Data accuracy
 - Spatial reasoning
 - Analytical judgment
 - Ethical map design

AI Policy

- Prohibited
 - Submitting AI-generated lab reports as original work
 - Using AI during in-class quizzes or practical exams
 - Generating fabricated data, citations, analysis, or map layouts
 - Without AI disclosure
- Permitted uses with disclosure
 - Clarifying GIS terminology
 - Debugging error messages
 - Brainstorming project ideas
 - Improving writing clarity in reports
 - Generating code examples for learning

Participation

- We will have pop quizzes!
- If you would like to ask a leave, you need to officially request via the myNTU system, and please email me and all TAs at the same time with the information below:
 - Department and year 系級
 - Name 姓名
 - Student ID 學號
 - Date 請假日期
 - Reason 請假事由

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